

Cover Letter to the Editor-in-Chief

Journal: International Journal of Refrigeration (Elsevier)

Date: September 7, 2026

Dear Editor-in-Chief,

Please find enclosed our original research manuscript titled **"Thermodynamic Phase-Equilibrium and Non-Linear Temperature Glide Modeling of Next-Generation Zeotropic A2L Refrigerants (R-454B & R-32)"** for consideration as a Research Article in the *International Journal of Refrigeration*.

With the global transition away from high-GWP HFCs (R-410A) mandated by the Kigali Amendment and EPA regulations, mildly flammable A2L refrigerants like R-454B (Opteon XL41) and R-32 represent the primary replacements across vapor-compression heating and cooling equipment. However, zeotropic blends like R-454B exhibit significant non-linear temperature glide during phase change. Traditional diagnostic practices using mid-point saturation approximations introduce systematic diagnostic errors (up to 2.2°F / 1.2°C) in subcooling and superheat evaluation.

In this paper, we formulate deterministic mathematical equations modeling liquid bubble and vapor dew curves directly from NIST REFPROP saturation boundaries, benchmarked against experimental field data. Furthermore, we evaluate the computational performance of running these formulations in client-side engineering software without external database dependencies.

We confirm that this work is original, has not been published previously, and is not currently under review elsewhere.

Sincerely,

Miad Saadidi

Lead Research Engineer

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